Jamie Fulford

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Education

University of Illinois Urbana-Champaign **August 2025 – Present**
Ph.D. in Computer Science *Champaign, Illinois*

- **Advisor:** Talia Ringer
- **Research interests:** programming languages, formal methods, interactive theorem proving, proof autoformalization

University of Virginia **August 2022 – May 2025**
B.S. in Computer Science and Mathematics *Charlottesville, Virginia*

- **GPA:** 3.9

Publications

Swarn Priya, Frédéric Besson, Connor Sughrue, Tim Steenvoorden, **Jamie Fulford**, Freek Verbeek, and Binoy Ravindran. BeePL: Correct-by-compilation kernel extensions. *arXiv preprint arXiv:2507.09883*, 2025.

Research Experience

University of Illinois Urbana-Champaign **August 2025 – Present**
Graduate Research Assistant, with Talia Ringer *Champaign, Illinois*

- **DARPA expMath:** building tooling for proof autoformalization and proof decomposition in Agda, targeting homotopy type theory. The tooling decomposes a theorem statement into lemmas that are dispatched to LLM-based provers.
- **Proof translation between program logics:** designing languages with specific interpretable properties and implementing them in Rocq, so that the languages interoperate and proofs compile between them.

Virginia Tech, Systems Software Research Group **January 2025 – July 2025**
Research Assistant (Proof Engineer) *Remote*

- Developed type systems and compiler verification in Rocq for BeePL, a language for safe Linux eBPF kernel extensions.
- Extended the CompCert verified compiler with a new frontend language.

University of Pennsylvania **May 2024 – August 2024**
NSF REU Research Fellow, with Steve Zdancewic *Philadelphia, Pennsylvania*

- Designed a lambda calculus for parallel and distributed computation and formalized it in Rocq, with machine-checked proofs of type safety and confluence.

Projects

Formalizing LLM Guardrails with AgentSpec in Lean 4 Paper Code **Fall 2025**

- Sole author. Executable Lean 4 semantics for AgentSpec, a DSL for runtime enforcement of LLM agent behavior, parameterized so the guarantees hold for any user-defined predicates.
- 42 machine-checked theorems with no `sorry`, including that safety is compositional in both directions, that the safety evaluator correctly implements the violation semantics, and that evaluation terminates.

ProbaLean: A Probabilistic DSL in Lean 4 Paper Code **Spring 2026**

- With Trey Plante and Zirui Zhou (equal contribution). A discrete probabilistic programming language in Lean 4 with an executable operational semantics and a denotational semantics over s-finite measures, built on Mathlib.
- Proved that the interpretation is a monad morphism into the Giry monad, that independent sampling commutes, and that the denotational semantics is sound with respect to the operational semantics.

Teaching Experience

Head Teaching Assistant, Computer Systems and Organization, University of Virginia (500+ students) **Fall 2023, Fall 2024**

Honors & Awards

PLMW Scholarship, Programming Languages Mentoring Workshop at POPL 2025 **January 2025**

Skills

Proof assistants: Agda, Rocq (Coq), Lean 4
Verification: Iris, CompCert, Mathlib
Programming languages: OCaml, Haskell, Rust, Python, C, C++, Java, JavaScript, SQL, Bash
Tools: Git, Linux, NixOS, LaTeX, PyTorch